

# A hematology-based clock derived from the Study of Longitudinal Aging in Mice to estimate biological age

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Biological clocks and other molecular biomarkers of aging are difficult to implement widely in a clinical setting. In this study, we used routinely collected hematological markers to develop an aging clock to predict blood age and determine whether the difference between predicted age and chronologic age (aging gap) is associated with advanced aging in mice. Data from 2,562 mice of both sexes and three strains were drawn from two longitudinal studies of aging. Eight hematological variables and two metabolic indices were collected longitudinally (12,010 observations). Blood age was predicted using a deep neural network. Blood age was significantly correlated with chronological age, and aging gap was positively associated with mortality risk and frailty. Platelets were identified as the strongest age predictor by the deep neural network. An aging clock based on routinely collected blood measures has the potential to provide a practical clinical tool to better understand individual variability in the aging process.

The aging process affects nearly all living organisms with many different manifestations, such as a decline in metabolic homeostasis, physical performance and cognitive abilities. Numerous factors across multiple domains differentially affect the pace of the aging process and highlight the fact that chronological age may not capture the diversity or accurately measure individual variation in the rate of biological aging. There is an emerging interest in estimating biological age using multi-omics strategies, including epigenetic markers of aging, which has resulted in the creation of different models or mathematical formulas to predict the age of an individual, broadly referred to as biological clocks. For

instance, epigenetic clocks based on DNA methylation (DNAm) accurately predict age and mortality risk<sup>1–3</sup>. DNAm clocks use the DNAm patterns at a specific number of loci based on thousands of samples at different ages. Using machine learning, these clocks can estimate biological age, remaining lifespan and a composite of age and physiological measurements known as ‘phenotypic indicators’, which were previously identified to predict mortality rates<sup>4</sup>. Researchers hypothesize that the discrepancies between the ages predicted using these biostatistical approaches and chronological ages represent the relative rate of aging, sometimes termed ‘age acceleration’ or ‘aging gap’. Such

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Table 1 | Sample size

| Study            | Cohort | B6 strain  |            |                | HET3 strain |            |                | DO strain  |            |                | Σ across sexes and strains |               |
|------------------|--------|------------|------------|----------------|-------------|------------|----------------|------------|------------|----------------|----------------------------|---------------|
|                  |        | F          | M          | Σ across sexes | F           | M          | Σ across sexes | F          | M          | Σ across sexes | Animals                    | Blood samples |
| SLAM             | C01    | 37         | 37         | <b>74</b>      | 36          | 33         | <b>69</b>      | –          | –          | –              | <b>143</b>                 | <b>571</b>    |
|                  | C02    | 35         | 38         | <b>73</b>      | 41          | 36         | <b>77</b>      | –          | –          | –              | <b>150</b>                 | <b>729</b>    |
|                  | C03    | 49         | 47         | <b>96</b>      | 49          | 46         | <b>95</b>      | –          | –          | –              | <b>191</b>                 | <b>979</b>    |
|                  | C04    | 50         | 47         | <b>97</b>      | 48          | 45         | <b>93</b>      | –          | –          | –              | <b>190</b>                 | <b>1,183</b>  |
|                  | C05    | 50         | 50         | <b>100</b>     | 50          | 48         | <b>98</b>      | –          | –          | –              | <b>198</b>                 | <b>1,021</b>  |
|                  | C06    | 59         | 56         | <b>115</b>     | 60          | 59         | <b>119</b>     | –          | –          | –              | <b>234</b>                 | <b>1,403</b>  |
|                  | C07    | 59         | 60         | <b>119</b>     | 62          | 60         | <b>122</b>     | –          | –          | –              | <b>241</b>                 | <b>1,275</b>  |
|                  | C08    | 50         | 47         | <b>97</b>      | 50          | 47         | <b>97</b>      | –          | –          | –              | <b>194</b>                 | <b>756</b>    |
|                  | C09    | 49         | 49         | <b>98</b>      | 57          | 62         | <b>119</b>     | –          | –          | –              | <b>217</b>                 | <b>1,183</b>  |
|                  | C10    | 60         | 58         | <b>118</b>     | 58          | 63         | <b>121</b>     | –          | –          | –              | <b>239</b>                 | <b>1,375</b>  |
| JAX              | G07    | –          | –          | –              | –           | –          | –              | 57         | 58         | <b>115</b>     | <b>115</b>                 | <b>305</b>    |
|                  | G08    | –          | –          | –              | –           | –          | –              | 56         | 60         | <b>116</b>     | <b>116</b>                 | <b>321</b>    |
|                  | G09    | –          | –          | –              | –           | –          | –              | 57         | 57         | <b>114</b>     | <b>114</b>                 | <b>298</b>    |
|                  | G10    | –          | –          | –              | –           | –          | –              | 57         | 50         | <b>107</b>     | <b>107</b>                 | <b>300</b>    |
|                  | G11    | –          | –          | –              | –           | –          | –              | 58         | 55         | <b>113</b>     | <b>113</b>                 | <b>311</b>    |
| Σ across cohorts |        | <b>498</b> | <b>489</b> | <b>987</b>     | <b>511</b>  | <b>499</b> | <b>1,010</b>   | <b>285</b> | <b>280</b> | <b>565</b>     | <b>2,562</b>               | <b>12,010</b> |

The dataset includes 12,010 samples from 2,562 mice (*M. musculus*) aged 11.6 weeks to 150.9 weeks, belonging to two studies—that is, the SLAM study, performed at the NIA, and the JAX longitudinal study of aging, whose data collection was executed from 2015 to 2023. It contains 15 cohorts (SLAM C01–C10 and JAX G07–G11), three strains (B6, HET3 and DO) and both sexes. The number of observations per animal ranged from 1 to 10 (average 4.8 samples per animal; s.d. 2.03). Bold numbers indicate the summation within a given row or column. Σ, summation; F, female; M, male.

measures can be used to identify aging biomarkers, uncover novel mechanisms, identify aging-specific population clusters and evaluate interventions aimed to slow or retard the aging process<sup>3</sup>.

Although epigenetic clocks are useful tools for aging research, DNAm data are expensive and computationally intensive to generate. Epigenetic clocks are, therefore, impractical for the long-term monitoring of changes in aging over time. A simple measure of the aging process that is affordable, easy to obtain and can be assessed more frequently over the entire lifespan may have considerable potential for the clinical application of biological clocks. For example, complete blood count (CBC) and fasting blood glucose (FBG) are among the most frequently ordered laboratory tests in biomedicine, and analysis of this type of data is common practice in the clinic<sup>5</sup>. The CBC, a widely used clinical assay in humans and mice, provides a wealth of information about the components of blood (for example, red blood cells (RBCs), white blood cells (WBCs) and platelets (PLTs)). A blood draw is a minimally invasive procedure routinely used in a clinical setting, and there is strong evidence of the interconnection between blood variables and aging progression<sup>6,7</sup>. For instance, functional changes in hematopoiesis, hemostasis and inflammation that occur via platelet–neutrophil interactions, fibrinolysis, anemia and elevated FBG have all been associated with poor outcomes and comorbidities, including an augmented risk of mortality<sup>8–12</sup>.

Prior research explored the potential of specific blood variables, in conjunction with other biomarkers, to assess the rate of aging and its potential impact on both mortality and morbidity in humans<sup>13–17</sup>. Body weight (BW) represents a metabolic parameter routinely evaluated in clinical practice that is closely linked to the development of age-related diseases. We recently examined the influence of BW, fat mass and FBG levels on mortality in mice, non-human primates and humans<sup>18</sup>, revealing the predictive capacity of BW as a marker of mortality, particularly during the later stages of life.

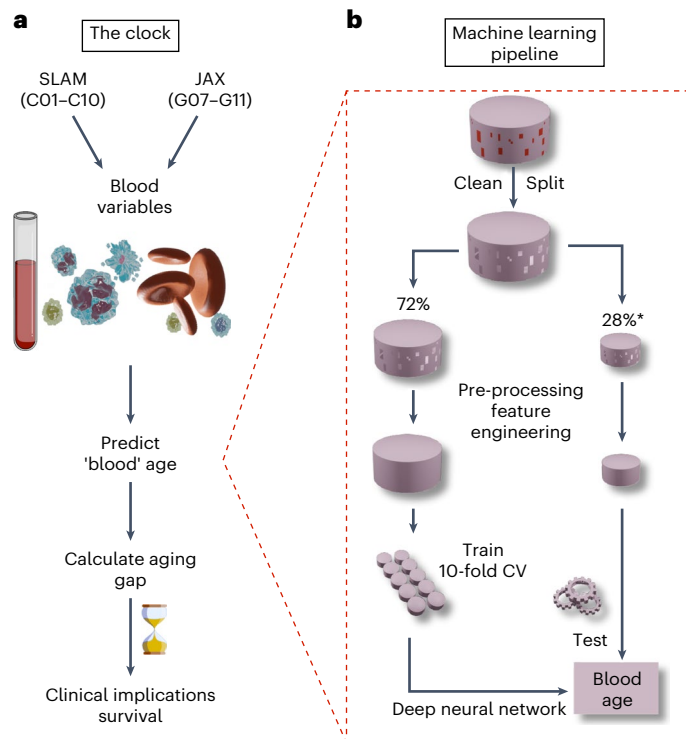
Mouse models in aging studies have the obvious advantages of short lifespan and genetic malleability. They provide the opportunity

to collect aging-related data without the interference of confounding variables or external exposures (for example, economic status, religion and education) that are perennially present in most longitudinal studies in humans, which can have a major impact on outcomes. Hence, the Study of Longitudinal Aging in Mice (SLAM) was launched in 2015 by the National Institute on Aging (NIA) to conduct studies aimed at better understanding the biology of aging in this commonly used preclinical animal model and address the challenge of finding clinically relevant metrics of biological aging<sup>19</sup>. Parallel longitudinal studies of mice are being carried out at other institutions, including The Jackson Laboratory (JAX). In the present study, we leveraged these two studies to assess whether fluctuations in blood variables can provide enough information to build a hematological clock capable of accurately determining biological age, using readily available biomarkers. This tool has the potential to be translatable to the clinic and provides an effective low-cost alternative to the need for highly specialized equipment and techniques required to use epigenetic clocks.

Results and Discussion

Data collection and study design

A machine learning pipeline, which constitutes the hematological clock, was designed to predict age in mice belonging to two independent studies. The cohort-based SLAM study consisted of 1,997 mice of both sexes and two different strains (that is, C57BL/6J (B6) and UM-HET3 (HET3)) that were housed at the NIA vivarium (Table 1; cohorts C01–C10). More than 10,475 blood draws were collected longitudinally with up to 10 samples per animal. Additionally, the JAX study consisted of 565 diversity outbred (DO) mice of both sexes, with up to three blood draws per animal for a total of 1,535 samples (Table 1; cohorts G07–G11). The DO mice were maintained on a different diet than the SLAM mice and were housed at the JAX animal facility. In all, our dataset contained 12,010 blood samples from 2,562 mice (Fig. 1a). A randomly selected cohort from each study (C07 and G08), accounting for roughly 10% of the total data, acted as one of the two testing sets (357 *M. musculus*



**Fig. 1 | Blood cell components as accurate predictors of biological age in mice.** **a**, Analysis workflow used to develop the hematological clock in B6 and HET3 (SLAM) and DO (JAX) mice of both sexes. **b**, Machine learning pipeline designed to predict blood age. After a cleaning process, the dataset was partitioned into two sets. One set was employed to train a DNN using a 10-fold CV approach, whereas the other was reserved for testing the model. 28%\*, held-out dataset encompassing randomly selected animals (18%) and randomly selected cohorts from the SLAM (C07) and JAX (G08) studies (~10%) accounting for approximately 28% of the initial dataset.

(177 females, 180 males, 116 DO, 119 B6 and 122 HET3) and 1,596 samples from 11.6 weeks to 130.4 weeks of age). The remaining mice were randomly divided 4:1 into a training set (~72%; 1,766 *M. musculus* (895 females, 871 males, 360 DO, 711 B6 and 695 HET3) and 8,365 samples from 12.0 weeks to 150.9 weeks of age) and a testing cohort (~18%) that, when combined with 'C07 G08', corresponded to the second testing set totaling 28% of the original mice labeled 'held-out' (796 *M. musculus* (399 females, 397 males, 205 DO, 276 B6 and 315 HET3) and 3,645 samples from 11.6 weeks to 145.6 weeks of age); (Fig. 1b). The training set (72%) served multiple purposes, including model selection, feature selection, hyperparameter tuning and training and validation of the final model. This comprehensive process was conducted using a 10-fold cross-validation (CV) methodology. The testing process was performed in the held-out set (28%) and within the randomly selected cohort C07 G08 to determine clock performance in a small independent group. All samples from an individual animal were included in the same dataset.

To identify the most effective model for predicting age, we evaluated three distinct algorithms—a deep neural network (DNN), a gradient boosting machine (GBM) and a generalized linear model (GLM)—incorporating Gaussian distribution and elastic net penalization. For initial model selection, overlapping hematological variables between the SLAM and JAX studies were included; however, for each pair of correlated variables that had a correlation coefficient (*R*) exceeding 0.80, one of the variables was systematically excluded. This initial set of 19 variables included RBC count, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), corpuscular hemoglobin concentration mean (CHCM), red blood cell distribution width (RDW), hemoglobin concentration distribution width (HDW), mean platelet volume (MPV),

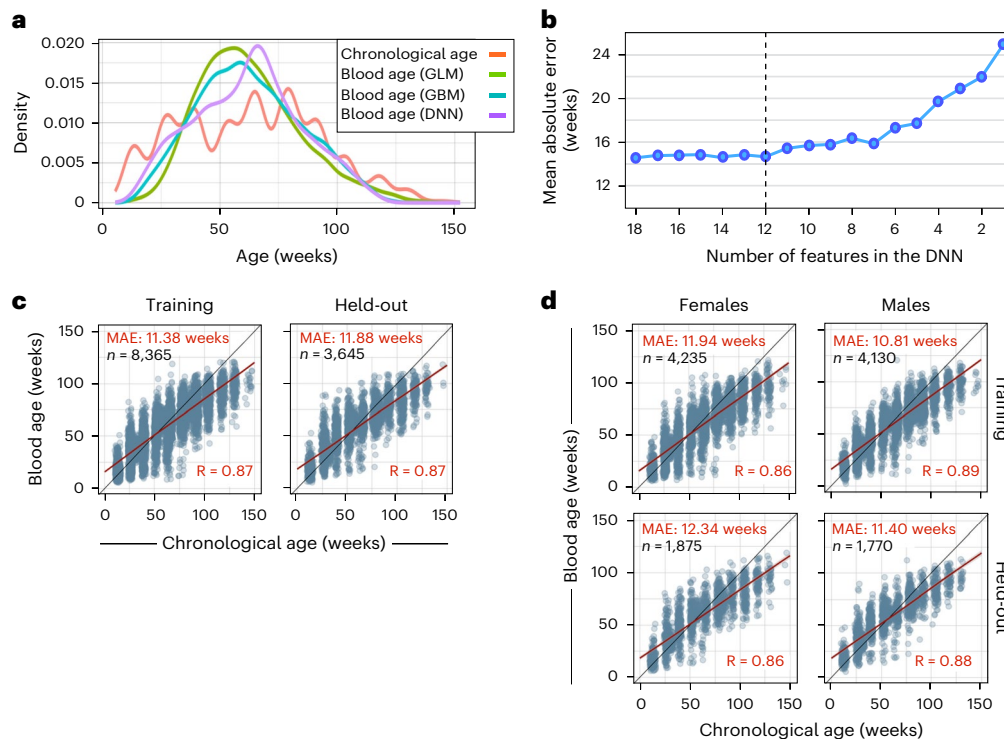
lymphocyte percentage (%LYM), eosinophil count (#EOS), eosinophil count percentage (%EOS), WBC count, neutrophil count (#NEU), monocyte percentage (%MONO), PLT count, FBG and BW. Sex and strain were also included in the models. The outcomes derived from the 10-fold CV process of these three models indicate that the DNN emerged as the most adept algorithm ( $MAE_{10FCV-DNN} = 14.36$  weeks,  $MAE_{10FCV-GBM} = 16.39$  weeks,  $MAE_{10FCV-GLM} = 14.61$  weeks) (Supplementary Table 1). Figure 2a shows the distribution of age predictions in the training set by each model compared to chronological age. In addition to the moderately improved performance of the DNN compared to other models, the increased accessibility of handling DNNs through platforms such as PyTorch<sup>20</sup> and H2O<sup>21</sup> makes processing and training times manageable, which provides advantages to incorporate a DNN in the hematological clock. Based on these considerations, the DNN was selected as the model of choice moving forward.

To assess whether the longitudinal structure of the data could provide further biologically relevant information and result in better prediction of blood age, the DNN was trained and tested by adding the longitudinal changes between timepoints for each blood variable. The inclusion of this information improved model performance ( $MAE_{10FCV-DNN} = 12.44$  weeks (with longitudinal changes and 19 features) versus  $MAE_{10FCV-DNN} = 14.36$  weeks (without longitudinal changes and 19 features)) (Supplementary Table 1).

### Optimal number of features in the DNN

To ascertain the optimal number of features for inclusion in the clock, we executed a feature selection process using the training set whereby the predictive performance of the model was compared after excluding one distinct independent feature at a time in the training dataset (Table 2). The sustained performance of the DNN model after retention of the top 12 most important features justified dropping the other features (Fig. 2b). Although the model's performance gradually deteriorated as more features were systematically removed from the set of features in the network, relatively good performance was still observed using only seven variables (Table 2). This indicates the applicability of the hematological clock across various experimental scenarios when some features may not be available. From this selection process, the final core set of features for the hematological clock consisted of 12 variables, comprising eight blood-related parameters (that is, RBC, MCV, MCH, CHCM, HDW, MCHC, PLT and WBC), two metabolic indices (BW and FBG) and demographic information, including sex and strain.

A hyperparameter tuning procedure using a grid search that combined different architectures, depths, number of neurons and regularization parameters was carried out through a 10-fold CV process, allowing the identification of an optimal configuration for the DNN (Supplementary Table 2) and subsequent testing of the model on unseen data. Using the core set of 12 features from the hematological clock, we then explored the relationship of chronological age with predicted blood age in the training dataset and the two testing subsets. The age prediction accuracy in the training set was comparatively similar with the predictions made in the two testing sets ( $MAE_{Training} = 11.38$  weeks,  $R_{Training} = 0.87$ ;  $MAE_{Held-out} = 11.88$  weeks,  $R_{Held-out} = 0.87$ ;  $MAE_{C07\ G08} = 12.05$  weeks,  $R_{C07\ G08} = 0.88$ ) (Fig. 2c and Extended Data Fig. 1a). To evaluate the possibility that these results were obtained by chance, we randomly permuted age as the dependent variable (50 iterations) while maintaining the values of all explanatory variables intact to provide a null distribution of statistics. Across 50 permutations, the models obtained average root mean square error (RMSE) of 34.12 weeks and mean absolute error (MAE) of 30.00 weeks, and none was below 32.31 weeks and 27.77 weeks, respectively (held-out set). The low correlations and larger RMSE and MAE values observed in this analysis indicated that the highly correlated predictions and small RMSE and MAE that we observed (Fig. 2c, Extended Data Fig. 1a and Supplementary Table 1) were unlikely to have been achieved by chance (Supplementary Table 3).



**Fig. 2 | Analyzing predictive performance of the hematological clock.**

**a**, Comparison of distributions of chronological age and blood age predicted by different models using blood samples from mice in the training set (1,766 mice, *M. musculus* (895 females, 871 males, 360 DO, 711 B6 and 695 HET3) and 8,365 samples from 12.0 weeks to 150.9 weeks of age). **b**, Predictive performance after feature removal from the DNN in the training set. **c**, Results of the clock's

predictions after processing samples from the training set and the 'held-out' testing set (796 mice, *M. musculus* (399 females, 397 males, 205 DO, 276 B6 and 315 HET3) and 3,645 samples from 11.6 weeks to 145.6 weeks of age). The red line shows an ordinary least squares (OLS) regression, with the shaded gray area indicating the 95% CI. Blue dots represent individual observations. **d**, Results of the clock's predictions stratified by sex. *n*, number of observations.

Stratified predictions based on sex in both the training and testing datasets revealed slightly higher correlations and lower errors between age prediction and chronological age in males compared to females (Fig. 2d and Extended Data Fig. 1b). The 'C07 G08' subset was used next to obtain stratified predictions based on three different age ranges—that is, less than 52 weeks (Young), between 52 weeks and 82 weeks (Adult) and more than 82 weeks (Old). The best performance in terms of correlation coefficient was obtained when the DNN predicted age using blood samples from old animals ( $R = 0.78$  (0.75–0.81)), then young ( $R = 0.68$  (0.65–0.71)) and finally adults ( $R = 0.49$  (0.46–0.52)), whereas the best behavior in terms of error was achieved from young mice (RMSE = 13.21, MAE = 9.36) (Extended Data Fig. 1c). The segregation of the 'C07 G08' testing set by strain showed a more accurate age prediction when computing samples from DO animals versus B6 and HET3 mice (Extended Data Fig. 1c), whereas a slightly better predictive performance was achieved when processing samples from the B6 strain alone compared to the clock prediction based on HET3 mice.

To explore the broader applicability of the clock, age predictions were evaluated in scenarios where specific variables were absent. Because the variables were standardized before being input into the clock, we conducted age predictions by censoring (that is, setting to 0) all possible combinations of variables, ranging from a single variable to half of the biomarkers in the held-out dataset (Supplementary Table 4). The predictive performance diminished substantially when variables were censored from the clock's core set of 10 biomarkers, especially the top-ranked variables in the feature importance process. The removal of the variable 'platelets' resulted in a more than five-fold increase in the MAE, whereas censoring the combination of the five variables 'MCHC', 'MCH', 'WBC', 'Glucose' and 'BW' led to less than two-fold increase in MAE compared to prediction that included the whole core set. Thus,

variables identified as central in the feature importance analysis played a pivotal role in maintaining the predictive accuracy of the clock.

Finally, we performed predictions by stratifying the held-out dataset into monthly age ranges to compare clock performance with cross-sectional data, excluding delta variables from the model (Supplementary Table 5). The model exhibits a comparatively stronger predictive performance on younger animals, demonstrating a relatively consistent trajectory until reaching 60 weeks of age, where the optimal performance was observed with an MAE of 7.12 weeks. However, interpreting the results from animals at the initial timepoint, specifically at 14 weeks of age (MAE = 19.20), poses challenges in drawing definitive conclusions.

We then examined the potential association between variability in blood components and clock performance in the three strains of mice through an intraclass correlation coefficient (ICC) analysis (Methods). Higher ICC was observed in most of the 10 features (hematological and metabolic) used in the clock when comparing the DO strain to either HET3 or B6 and between HET3 versus B6 (Supplementary Table 6). The stratified predictions indicate clear strain-specific differences in the predicting performance of the clock, which were best in the DO stock (Extended Data Fig. 1c). In a previous report, genetic heterogeneity<sup>22</sup> was proposed to contribute to higher CBC variability in DO mice<sup>23</sup>, which may have had some role in the superior clock performance in this strain. However, the heterogeneous HET3 mice had higher CBC variability but less accurate predictions than inbred B6 mice, making it difficult to ascertain any correlation among genetic diversity, age-associated blood dynamics (for example, CBC variability) and clock performance. The degree of dissimilarity in the ICCs between strains suggests that environmental factors, rather than genetic diversity, may have played a dominant role in shaping the aging patterns captured by the clock. Another possible rationale for the enhanced predictive accuracy in DO mice lies in the substantial genetic influence on age, which could



**Table 2 | Decline in DNN performance for blood age estimation by eliminating relevant features through a 10-fold CV process using the training dataset (72% data)**

| #Feat | MAE   | RMSE  | R    | Features in the DNN                                                                      | #Out |
|-------|-------|-------|------|------------------------------------------------------------------------------------------|------|
| 18    | 14.56 | 18.77 | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,mpv,p.lym,n.eos,wbc,n.neuts,p.eos,p.mono,plt,fgb,bw | 1    |
| 17    | 14.78 | 19.08 | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,mpv,p.lymp,n.eos,wbc,n.neuts,p.mono,plt,fgb,bw      | 2    |
| 16    | 14.81 | 19.11 | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,mpv,p.lymp,n.eos,wbc,n.neuts,plt,fgb,bw             | 3    |
| 15    | 14.85 | 19.06 | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,mpv,p.lymp,wbc,n.neuts,plt,fgb,bw                   | 4    |
| 14    | 14.63 | 18.83 | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,mpv,p.lymp,wbc,plt,fgb,bw                           | 5    |
| 13    | 14.82 | 19.1  | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,p.lymp,wbc,plt,fgb,bw                               | 6    |
| 12    | 14.71 | 19.01 | 0.77 | strain,sex,rbc,mcv,mch,mchc,chcm,hdw,wbc,plt,fgb,bw                                      | 7    |
| 11    | 15.45 | 19.78 | 0.75 | strain,sex,rbc,mch,mchc,chcm,hdw,wbc,plt,fgb,bw                                          | 8    |
| 10    | 15.72 | 20.09 | 0.74 | strain,sex,rbc,mch,mchc,chcm,hdw,plt,fgb,bw                                              | 9    |
| 9     | 15.76 | 20.1  | 0.74 | strain,sex,rbc,mch,mchc,chcm,hdw,plt,bw                                                  | 10   |
| 8     | 16.37 | 20.73 | 0.72 | strain,sex,rbc,mch,mchc,chcm,plt,bw                                                      | 11   |
| 7     | 15.86 | 20.2  | 0.73 | strain,sex,rbc,mch,chcm,plt,bw                                                           | 12   |
| 6     | 17.34 | 22.23 | 0.66 | strain,sex,rbc,chcm,plt,bw                                                               | 13   |
| 5     | 17.72 | 22.71 | 0.64 | strain,sex,rbc,plt,bw                                                                    | 14   |
| 4     | 19.74 | 25.17 | 0.53 | strain,sex,plt,bw                                                                        | 15   |
| 3     | 20.91 | 26.04 | 0.48 | strain,sex,plt                                                                           | 16   |
| 2     | 21.97 | 27.12 | 0.41 | sex,plt                                                                                  | 17   |
| 1     | 24.97 | 30.06 | 0.16 | sex                                                                                      | 18   |

#Feat, number of features in the DNN; #Out, number of features removed from the DNN; bw, body weight; chcm, corpuscular hemoglobin concentration mean; fgb, fasting blood glucose; hdw, hemoglobin concentration distribution width; mch, mean corpuscular hemoglobin; mchc, mean corpuscular hemoglobin concentration; mcv, mean corpuscular volume; mpv, mean platelet volume; n.eos, number of circulating eosinophils; n.neu, number of circulating neutrophils; p.eos, eosinophils (%); p.lym, lymphocytes (%); plt, platelets; rbc, red blood cell count; rdw, red blood cell distribution width; wbc, white blood cell count.

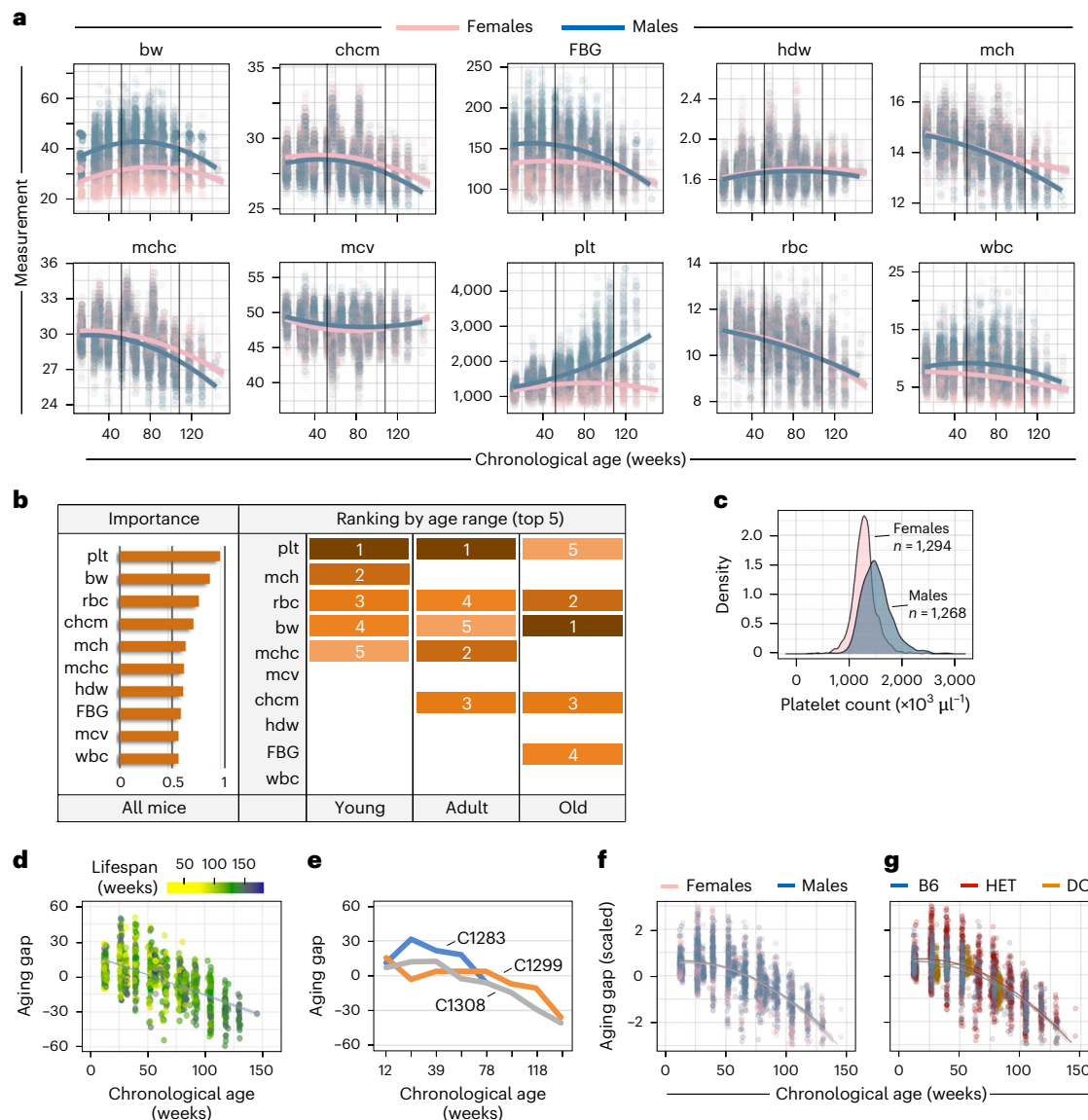
imply a comparatively diminished stochastic component. A third, non-biological factor that could confer better performance of the clock in DO mice is the longitudinal structure of the samples. The JAX study includes blood samples from three specific timepoints (that is, 12 months, 18 months and 24 months) as opposed to the SLAM study, which comprises samples from up to 10 timepoints. Hence, the clock is required to predict a more complex outcome in the SLAM study. We took several measures to address the vulnerability of models such as DNNs to overfitting<sup>24</sup> by avoiding information leakage between datasets, not overtraining the network and including diverse penalization techniques, such as dropout or elastic net regularization, among others (Methods). The similar clock predictions in unseen blood samples from the two testing sets compared to the training set suggest appropriate behavior of the model regarding overfitting.

Because methylation-based clocks have demonstrated their effectiveness in predicting chronologic age<sup>1–3,5,16</sup>, we carried out a comparative analysis in the performance of our hematological clock with established models to yield valuable insights into its applicability and versatility. A previous investigation by Stubbs et al.<sup>25</sup> tailored a methylation-based age predictor specifically for mice and subsequently compared its accuracy with that of the 175 Horvath clock regions designed for humans. The authors reported a correlation between predicted and chronological age in the validation dataset of  $R = 0.84$ , which falls well within the correlation of  $R = 0.88$  in the ‘C07 G08’ testing set using blood variables. A decrease in the accuracy of age prediction was observed when applying methylation data from mice to the Horvath human clock<sup>1</sup> compared to their mouse-specific clock. In another study conducted by Thompson et al.<sup>26</sup>, the authors devised four distinct epigenetic clocks for mice through the selection of diverse penalized regression models, the incorporation of various CpG sets and the use of different validation techniques<sup>26</sup>. Once again, the performance of these clocks exhibited correlation coefficients spanning from  $R = 0.64$  to  $R = 0.82$  when assessed on the held-out dataset.

To conclude, similar predicting performance of our DNN-based hematological clock was obtained at two sites, three strains and both sexes. Adequate clock behavior was observed in terms of overfitting, and the comparison with a gold standard methylation clock indicates an equivalent ability to predict age. This further underscores the fact that the hematological clock, relying on just 12 readily accessible features (that is, 10 biological factors, sex and strain), serves as a viable and low-cost alternative to other models, which can be easily implemented across different laboratories. This not only will advance the field of aging research but could also be used in other age-related settings, including drug or intervention testing.

**Longitudinal feature importance**

From the 10 biological features included in the clock, some of them displayed sexual dimorphism (Fig. 3a), which might influence the predictive ability of the clock. To investigate this issue further, we initially determined the relevance of each feature within this core set of blood variables and assessed whether the ranking of importance remained constant with age by performing a feature importance analysis and stratifying the entire dataset (12,010 samples) in three age groups: young (tercile 1 age (12–40 weeks)); adult (tercile 2 age (41–78 weeks)); and old (tercile 3 age (79–151 weeks)), with 4,004 samples (1,708 B6, 512 DO and 1,784 HET3) in terciles 1 and 2 and 4,002 samples (1,707 B6, 511 DO and 1,784 HET3) in tercile 3. We applied a method based on performance decline in absence of features within a DNN CV process<sup>27</sup> to determine the importance of these features, which were found to be uneven throughout life (Fig. 3b). PLT count (adjusted by clumps) was the most important feature, appearing as a strong predictor of age during the young and adult stages, with diminished relevance in the old animals. Viewed as the second most important feature predicting blood age, BW gained the most relevance for the model in late adulthood (tercile 3 age). A similar pattern was observed for RBC count, whereas FBG had a relatively low



**Fig. 3 | Analyzing feature importance and aging gap. a**, Sex-stratified trajectories of eight blood variables and two metabolic indices used as features in the hematological clock from B6 and HET3 (SLAM) and DO (JAX) mice of both sexes (2,562 mice, *M. musculus* (1,294 females, 1,268 males, 565 DO, 987 B6 and 1,010 HET3) and 12,010 samples from 11.6 weeks to 150.9 weeks of age). The pink and blue lines represent smoothed trends of each sex, generated using the LOESS method, with the shaded gray areas representing the 95% CI for the estimated trend and the dots representing individual observations. bw, body weight; chcm, corpuscular hemoglobin concentration mean; hdw, hemoglobin distribution width; mch, mean corpuscular hemoglobin; mchc, mean corpuscular hemoglobin concentration; mcv, mean corpuscular volume; plt, platelet count; rbc, red blood cells; wbc, white blood cells. **b**, Feature importance assigned by the DNN applying the Gedeon et al. method<sup>27</sup> during a 10-fold CV process and after processing blood samples from all mice in the study (left panel) (2,562 mice, *M. musculus* (1,294 females, 1,268 males, 565 DO, 987 B6 and 1,010 HET3) and 12,010 samples from 11.6 weeks to 150.9 weeks of age) and from the same animals segregated by age (right panel). **c**, Distribution of average PLT counts per mouse, stratified by sex, including all animals from both

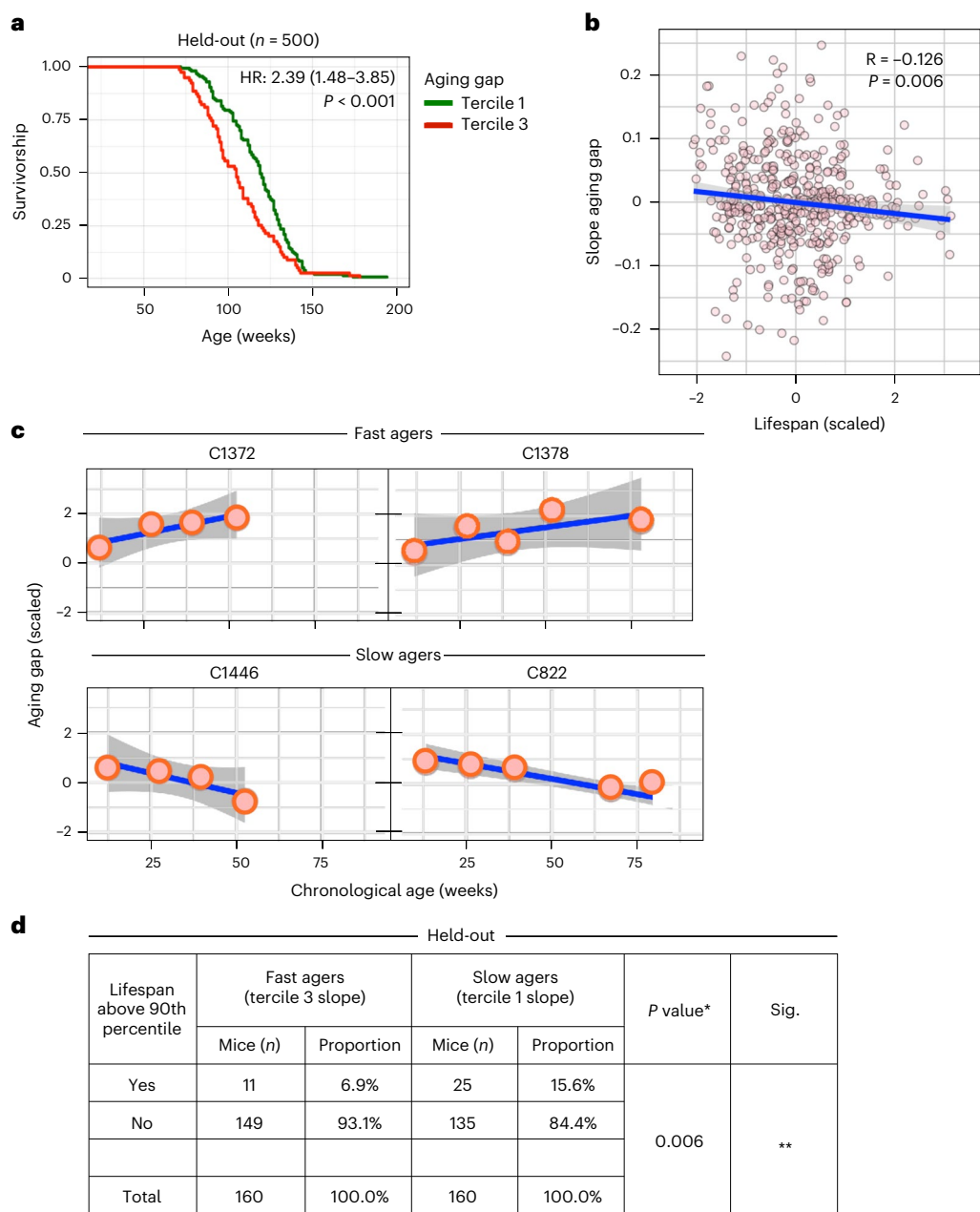
the SLAM and JAX studies. Only blood samples collected up to 84 weeks of age, the final timepoint for DO mice in the JAX study, are included (2,562 mice, *M. musculus* (1,294 females, 1,268 males, 565 DO, 987 B6 and 1,010 HET3) and 9,485 samples from 11.6 weeks to 84.0 weeks of age). **d**, Relationship between chronological age on clock-derived aging gap in the held-out set (796 mice, *M. musculus* (399 females, 397 males, 205 DO, 276 B6 and 315 HET3) and 3,645 samples from 11.6 weeks to 145.6 weeks of age). The blue line represents an OLS regression; the shaded gray area represents the CIs at 95%; and the yellow to violet colored dots (shorter to longer lifespan of each mice, respectively) represent individual observations. **e**, Connected scatterplot showing the aging gap trajectories over the lifespan of three representative mice in the study. **f,g**, Aging gap trajectories (scaled) over the lifespan of mice after stratification by sex (**f**) and strain (**g**) in the held-out set (796 mice, *M. musculus* (399 females, 397 males, 205 DO, 276 B6 and 315 HET3) and 3,645 samples from 11.6 weeks to 145.6 weeks of age). The lines represent the smoothed trajectories (LOESS method) of each sex or strain; the shaded gray areas represent the CIs at 95%; and the dots represent individual observations. LOESS, locally estimated scatterplot smoothing.

age-associated predictive power during the latter part of life (Fig. 3b and Supplementary Table 7).

Notably, the average PLT counts were significantly higher in males than in females independent of the study and strain of mice (males versus females  $P < 0.001$ : B6,  $1,585 \pm 244$  versus  $1,330 \pm 191$ ; HET,  $1,372 \pm 197$  versus  $1,215 \pm 146$ ; DO,  $1,650 \pm 374$  versus  $1,344 \pm 313$  (units:  $\times 10^3 \mu\text{l}^{-1}$ ))

(Fig. 3c). Moreover, the longitudinal BW and FBG trajectories were higher in males than in females, whereas the trajectory of the remaining variables did not show clear sex differences (Fig. 3a).

The DNN-based model applied in this study detects and prioritizes features that exhibit substantial changes throughout the mouse lifespan. The features with a more drastic pace of change held top importance



**Fig. 4 | Analyzing the impact of aging gap on survivorship in the 70-week window.** **a**, Survival analysis, categorical: Kaplan–Meier curves depict mice categorized as fast agers (tercile 3 aging gap) and slow agers (tercile 1 aging gap) across the population in the held-out dataset alive after 70 weeks (500 mice, *M. musculus* (253 females, 247 males, 164 DO, 163 B6 and 173 HET3) and 1,529 samples from 11.6 weeks to 70 weeks of age) and considering aging gaps from baseline to 70 weeks of age only (70-week window). The aging gap distribution within the 70-week window was segregated into terciles, and the hazard ratios were estimated using recurrent extended Cox models, considering aging gap as a categorical time-dependent covariate and computing the duration each mouse spent in each tercile of the aging gap distribution. **b**, Relationship between slope aging gap and lifespan (scaled). A linear mixed-effect model was used to extract individual aging gap slopes in the 70-week window and then test for the association between slope aging gap and lifespan using linear regression. Only samples from animals alive after 70 weeks were used to perform this analysis (480 mice, *M. musculus* (243 females, 237 males, 150 DO, 160 B6 and

170 HET3)). The blue line shows an OLS regression, with 95% CI, accompanied by the correlation coefficient,  $R$ , and the associated  $P$  value (two-sided  $t$ -test). Pink dots represent individual slopes. **c**, Aging gap trajectories across life of four representative mice in the held-out dataset: two fast agers according to their slopes (tercile 3 aging gap slope, top panel) and two slow agers (tercile 1 aging gap slope, bottom panel). The blue lines represent the best-fit straight line; the shaded gray areas represent the CIs at 95%; and the red dots represent individual observations of a given animal. **d**, Results of the maximum lifespan comparison<sup>42</sup> show the proportion of animals living above the 90th percentile lifespan (144 weeks) in the fast and slow aging groups based on the slope aging gap categorization. This analysis includes samples from baseline to 70 weeks, drawn from animals in the held-out dataset with at least two timepoints: instance of natural death and alive after 70 weeks (480 mice, *M. musculus* (243 females, 237 males, 150 DO, 160 B6 and 170 HET3)).  $P$  values were calculated using a two-sided Boschloo’s test. HR, hazard ratio; OLS, ordinary least squares; Sig., significance.

rankings, and sexual dimorphism seems not to have been a decisive factor in determining the importance attributed to many of these variables. PLT count is directly associated with aging through mechanisms such as

hemostasis, coagulation and inflammation<sup>28,29</sup>. Notably, the high rank assigned by the model to BW further validates previous findings regarding its significance as a predictor of age and its implication in mortality<sup>18</sup>.

Although studies of epigenetic clocks demonstrated the value of predicting biological age based on a single sample/timepoint, the data from our hematological clock demonstrate that the aging gap can vary within the same animal over time (Fig. 3d,e), which could pose challenges for the interpretability of cross-sectional measurements using our hematological clock. The collection of longitudinal data enables age prediction and the calculation of aging gap at various timepoints, providing a more complete picture of the rates of biological aging across time.

### Aging gap and its association with lifespan

Aging gap was calculated as the difference between blood age (predicted by the clock) and chronological age and then examined for its potential clinical implications in two key outcomes: mortality and frailty.

The well-established influence of sex and genetic background (for example, strain) on mouse aging<sup>30–32</sup> led us to assess their impact on aging gap. There was a significantly higher aging gap among males compared to females in the held-out dataset ( $\beta_{\text{males}} = 1.1$ ,  $t = 2.43$ ,  $P = 0.02$ ) (Fig. 3f) combined with higher aging gap in HET3 relative to B6 mice ( $\beta_{\text{HET3}} = 2.9$ ,  $t = 6.3$ ,  $P \leq 0.001$ ) but not in DO compared to the other two strains (Fig. 3g).

One of the most intriguing findings from this work is the high degree of heteroskedasticity in the model's prediction errors—that is, aging gap—with old mice appearing younger (low aging gap) and young mice appearing somewhat older (high aging gap) (Fig. 3f,g). This observation may indicate the presence of survival bias. To investigate this, we conducted a survival analysis using Cox regression<sup>33</sup>, focusing only on data collected before 70 weeks of age and excluding any animals that died before this point (Fig. 4a). The stringent restriction of the data used for these predictions was done to both ensure balanced representation of the groups across the period being assessed and discard eventual result bias due to the observed heteroskedasticity. This approach aims to determine whether mice that show slower aging before most of deaths occur tend to live longer than mice of the same chronological age that are aging more rapidly.

Assaying mortality using Cox regressions by modeling time to death as the outcome and aging gap as a time-dependent exposure (that is, a covariate that changes at each timepoint) showed higher mortality risk in those mice with higher aging gap ( $\text{HR}_{\text{held-out-T0-70-week}} = 1.03$  (1.01–1.04);  $P < 0.001$ , confidence interval (CI) = 0.56, events = 500). Indeed, 'held-out' mice within the highest tercile for aging gap had a significantly higher mortality rate than those within the lowest tercile, after adjustment for cohort, strain, sex and age ( $\text{HR}_{\text{held-out-T3vsT1-70-week}} = 2.39$  (1.48–3.85),  $P < 0.001$ , CI = 0.55, events = 500) (Fig. 4a). The results obtained in the Cox survival analyses were further evaluated by fitting Gompertz regressions to the data and comparing subpopulation survivorship. This second approach also identified greater mortality risk within the animals with higher aging gap ( $\text{HR}_{\text{held-out-T3vsT1-70-week}} = 1.90$  (1.42–2.55),  $P < 0.001$ , events = 500).

To determine whether specific aging gap patterns could be linked to an individual's lifespan, we analyzed the slopes of the aging gap trajectories in the 70-week window and their relationship with lifespan from animals that were unseen during construction of the clock. The results showed that animals still living at 70 weeks of age with steeper aging gap slopes lived shorter compared to those with shallower slopes (Fig. 4b), indicating a negative association between the aging gap slopes earlier in life and individual lifespans. We then used a different analytical approach whereby the slopes in the 70-week window were categorized into terciles, and the proportion of individuals living beyond the P90 lifespan (144 weeks in the held-out group) were compared between the bottom and top terciles, defined as slow agers (T1 aging gap slope) and fast agers (T3 aging gap slope) (Fig. 4c,d). The trajectories of four mice ranked as fast or slow agers according to their slopes are provided as an example (Fig. 4c). The fast agers had a

notably lower proportion of individuals older than 144 weeks (6.9%) compared to their slow ager counterparts (15.6%) ( $P = 0.006$ , two-sided Boschloo's test) (Fig. 4d).

Each of the analyses conducted using blood samples from the held-out dataset showed significant correlation between the determined aging gap and lifespan. Here, animals were classified as slow or fast agers during their youth or adulthood and then tested for longevity, yielding significant results. Further analyses including all animals in the training and testing datasets, without limiting the observations to the 70-week window, confirm our findings and showed consistency across datasets (Supplementary Table 8 for stratified results and Extended Data Fig. 2). Our trajectory-based approach holds particular importance owing to the considerable interest in identifying modifiable aging trajectories early in life to investigate whether specific behavioral, dietary and/or pharmacological interventions can influence these trajectories, ultimately impacting the lifespan of a particular individual.

One important aspect affecting the results presented in this study is the euthanasia process performed in the cross-sectional arm of the SLAM study to create a tissue bank. From the longitudinal arm of SLAM, only animals that died of natural death were included in the analyses involving lifespan as an outcome or as a predictor (Supplementary Table 9). Thus, this dataset lacks lifespan-related information from animals euthanized for specimen collection that could have been long lived, which reduces the average maximum lifespan of the entire population. This information must be considered during the interpretation of the results from the analyses concerning lifespan and mortality, especially for females whose average life expectancy is generally longer than for males. Nevertheless, this approach provided corroborating evidence of an association between longitudinal patterns of blood components and lifespan in this preclinical model.

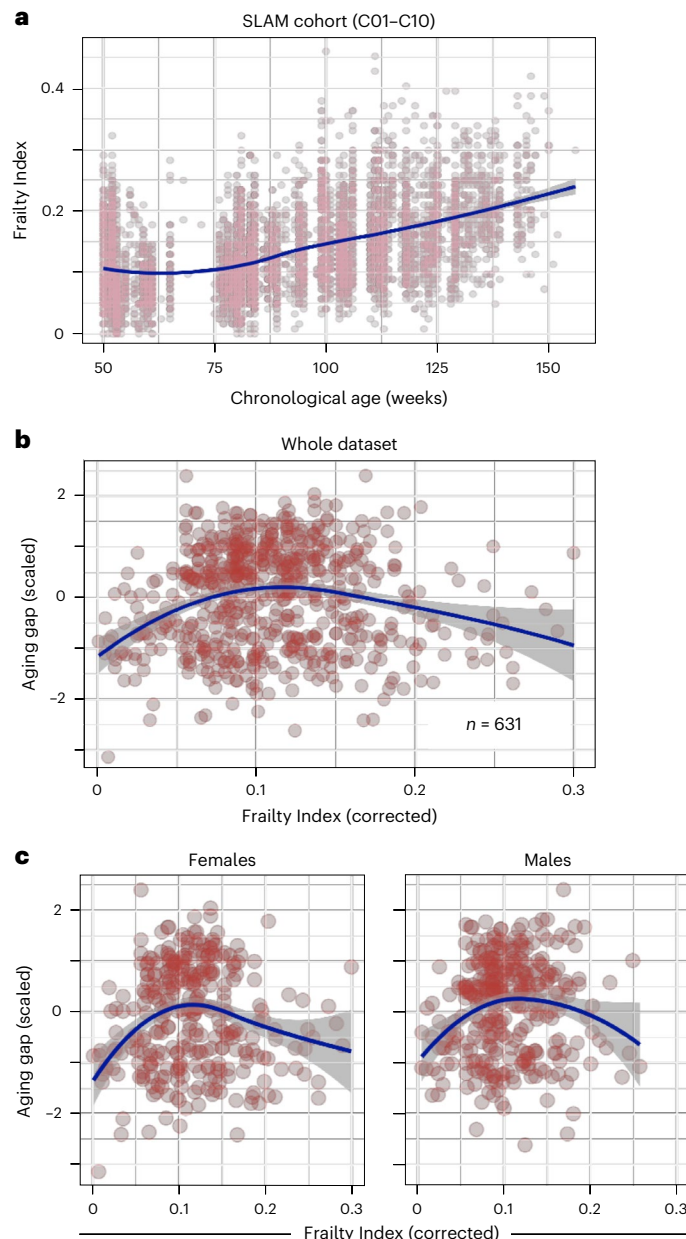
### Higher aging gap is associated with frailty in older mice

Assessment of the Frailty Index (FI) allows researchers to longitudinally monitor age-associated phenotypic changes that are strongly associated with mortality and morbidity<sup>34–37</sup>. This index, based on a collection of 31 health deficits directly linked to age, was obtained from the SLAM cohort mice (C01–C10) (Fig. 5a) that had undergone blood draws, resulting in a collection of 631 FI scores from 562 mice. Considering the substantial impact of the operator on the scoring process and given that all indexes were produced by a consistent group of seven highly trained individuals, we sought to detect and correct the individual influence of each evaluator on the FI scores (Methods).

Linear mixed-effects regression analyses showed a clear correlation between aging gap and FI ( $\beta_{\text{aging gap}} = 0.13$ ,  $t = 3.29$ ,  $P = 0.001$ ), both in females ( $\beta_{\text{aging gap-F}} = 0.13$ ,  $t = 2.76$ ,  $P = 0.006$ ) and in males ( $\beta_{\text{aging gap-M}} = 0.11$ ,  $t = 1.993$ ,  $P = 0.047$ ). To account for the nonlinear relationship between these two variables, we fitted a second model encompassing a piecewise regression with a breaking point in the FI minimizing the model's mean standard error among all models tested (FI = 0.90). We observed that the FI was positively associated with aging gap in both regressions ( $t_1 = 4.76$ ,  $P < 0.001$ ,  $t_2 = 3.50$ ,  $P < 0.001$ ) and in the first regression when stratifying by sex ( $t_{1M} = 3.60$ ,  $P < 0.001$ ,  $t_{2M} = 1.81$ ,  $P < 0.07$ ,  $t_{1F} = 3.55$ ,  $P < 0.001$ ,  $t_{2F} = 1.90$ ,  $P < 0.06$ ). Thus, fast aging animals had a greater prevalence of frailty than those with lower aging gap regardless of sex (Fig. 5b,c).

We conducted two additional analyses to evaluate the ability of the blood-derived aging gap to differentiate young individuals with unusually high FIs and older individuals with unusually low FIs, from the remaining population. SLAM mice that fell in the lowest/highest tercile of the age distribution with the highest/lowest FI (in the upper/lower tercile of the FI distribution, adjusted for operator and batch effects) were identified, and logistic regression was used to evaluate the relationship between the frequencies of these subsets of animals (that is, Young High Frailty (YHF) (78 *M. musculus*, 35 females and 43





**Fig. 5 | Association between the FI and aging gap estimate.** **a**, Rate of change in frailty in the SLAM mice (cohorts C01–C10). Each point represents one FI raw measure, and the line represents a smoothed trend generated using the LOESS method with 95% CI (2,172 mice, *M. musculus* (1,085 females, 1,087 males, 1,065 B6 and 1,107 HET3) and 5,023 samples from 50.0 weeks to 156.0 weeks of age). **b**, Relationship between aging gap and FI adjusted for operator and batch and smoothed by the LOESS method with 95% CI. Each point represents an FI measure after eliminating the operator effect and the cohort effect using linear mixed-effect models (Methods). **c**, Relationship between aging gap and FI (LOESS method) with 95% CI, adjusted for operator and batch, stratified by sex. **b,c**, Generated from SLAM animals with blood measures and FI (562 mice, *M. musculus* with blood measures and FI scores (278 females, 284 males, 239 B6 and 323 HET3) and 631 samples from 51.9 weeks to 143.4 weeks of age). LOESS, locally estimated scatterplot smoothing.

males, 29 B6 and 49 HET3, from 51.9 weeks to 79.0 weeks of age) and Old Low Frailty (OLF)) (63 *M. musculus*, 26 females and 37 males, 21 B6 and 42 HET3, from 117.4 weeks to 143.4 weeks of age) and aging gap.

For each unit increase in aging gap, there was a 30% rise in the odds of being classified as YHF after accounting for available covariates ( $P < 2 \times 10^{-16}$ ) and a 26% decrease in the odds of falling into the OLF

category ( $P = 1.01 \times 10^{-5}$ ). This suggested that the aging gap variable plays a crucial role in discerning individuals of the same chronological age based on their degree of frailty. Thus, higher aging gaps are associated with an increased likelihood of being classified as YHF while decreasing the odds of being categorized as OLF. Considering that frailty serves as a measure of vulnerability to adverse health outcomes, this finding highlights the clock's ability to discern mice of the same chronological age that have different physiological status. This holds importance due to the heightened risks that frail individuals face, including increased mortality rates, greater susceptibility to various diseases and disorders, potential diminished response to treatments or interventions and a potentially worse prognosis when confronted with the same illness compared to non-frail individuals<sup>36,37</sup>.

More than 12,000 observations on approximately 2,500 mice were used to build and validate the hematological clock. It is a highly controlled study with a good representation of mice at different ages, based on three different strains and both sexes. By considering not only blood cell counts at specific timepoints but also the longitudinal changes across time, this method is able to measure the aging process and transform the set of blood variables into a biologically meaningful metric of aging. Used as the clock's output, aging gaps were computed for each individual blood collection, and a single longitudinal trajectory per animal was generated. An important aspect of the clock's predictive capability for age lies in the fact that the hematological clock extends beyond the mere assessment of individual parameters and/or cellular components. Instead, it directly quantifies blood cell populations. This approach could offer a potential advantage in capturing variations in essential biological mechanisms linked to the pace of aging.

As a final consideration, several interesting avenues of research are raised by this work beyond the need of translating this method to humans. These include changes in aging gap trajectories in response to interventions that either shorten lifespan (for example, radiation and high-fat diets) or extend lifespan (for example, caloric restriction, rapamycin and senolytic drugs). Examination of the hematological clock's behavior in heterochronic and isochronic parabiotic pairings and in mice with specific mutations that promote blood-related diseases, such as leukemia or anemia, would also be of great interest. Further data mining could also be informative—for example, using the whole raw ADVIA output with more than 500 variables (not all of them biologically relevant) to determine which features are better predictors of age and whether other sets of variables outperform the one evaluated in this work. Moreover, machine learning tools could be helpful in identifying putative interactions among these and other less studied variables as well as predicting different time-to-event outcomes other than mortality, such as tumor onset or the inception of cognitive-associated morbidities.

This study has several strengths and limitations that are worth noting. As the first large longitudinal study of normative aging in mice, SLAM assesses many phenotypes, biological metrics, physical variables and pathologies, which results in a very rich dataset with a large sample size. Unlike the data used to generate other aging clocks, the blood features used in this analysis are routinely collected in a clinical setting and could provide a unique opportunity to assess aging progression in human populations. Our analysis used data from two different strains of mice of both sexes, including both inbred and genetically heterogeneous populations of mice and data from a study conducted at a different facility in an even more genetically diverse mouse population, further emphasizing the robustness of our findings. Although including strain as a predictor restricts the generalizability of the DNN to only three strains, we think that it enhances the robustness of the mortality aspect of this study. This choice, although potentially limiting the tool's applicability to other strains, contributes substantially to improving the reliability of the study's findings.

One clear limitation of this study stems from the planned euthanasia process for the tissue bank construction, which has a direct

impact on mortality and lifespan. Besides the obvious highly controlled environment of our animal facilities, another limitation arises from the fact that the animals in the study were virgin mice and housed at sub-thermoneutral conditions, which is different from what would be observed in human populations. Another potential limitation to our analysis is the possibility of overfitting, as our DNN model has a multitude of parameters relative to the sample size. Ideally, we would validate our results in an external dataset; however, this is unfeasible in practice, as these are the only data of this kind. We have taken steps to select our validation data in ways that ensure a fair evaluation of the model, and we used several techniques to mitigate overfitting. We have no evidence of substantial overfitting, but future studies could shed light on this issue. In the current configuration of the clock, it is possible for the model to infer that DO mice originate from JAX, where mice are aged between 12 months and 24 months, whereas B6 and HET3 mice may encompass ages outside of this range. Notably, this discrepancy is unrelated to genetic or biological distinctions among the strains, which was the primary rationale for including strain as a variable. This could potentially impact the accuracy observed in the reported results and likely decrease accuracy when extrapolating findings to strains with varying study designs.

In conclusion, this model can serve as a viable alternative to epigenetic clocks and a valuable tool to help drive forward the aging field. We demonstrated that a biological clock based on routinely collected blood variables can provide reliable predictions of age. The difference between blood age and chronological age appears to be associated with mortality risk, as mice with higher predicted age versus chronological age have nearly a 3.2-fold higher death rate. Furthermore, we identified a clear association between aging gap and frailty across sexes. Translation of these findings to clinical application in humans will require further study to account for the myriads of uncontrolled variables and challenges that study participants face daily.

## Methods

This research complies with all relevant ethical regulations. The SLAM study was approved by the Animal Care and Use Committee of the NIA under Animal Study Protocol TGB-458-2022. All procedures and protocols related to the DO study were reviewed and approved by the JAX Institutional Animal Care and Use Committee (IACUC) under protocol IACUC-06005 (Jackson Center for Aging Research). The study was carried out in accordance with National Institutes of Health Guidelines for the Care and Use of Laboratory Animals.

### SLAM

The SLAM study is an ongoing, longitudinal cohort-based assessment of normative aging in mice (*M. musculus*) conducted at the NIA<sup>19</sup>. Inbred B6 (stock no. 000664) and outbred HET3 (stock no. 036603) mice of both sexes were born at JAX, shipped to the NIA animal facility at 12 weeks of age in cohorts of approximately 200 animals every 3 months and followed longitudinally thereafter. For the current analysis, data from cohorts 1–10 (~50% female and ~50% HET3) were collected between 2015 and 2023 (Table 1). Mice were group housed (four mice per cage) and fed an NIH-31 diet ad libitum with free access to water. Mice were housed in rooms maintained at  $22.2 \pm 1^\circ\text{C}$  and 30–70% humidity. Routine tests were performed to ensure that mice were pathogen free, and sentinel cages were maintained and tested according to American Association for Accreditation of Laboratory Animal Care criteria.

To enhance the generalizability of our hematological clock, we incorporated data on blood variables, FBG and BW from a cohort of DO mice (*M. musculus*) previously described in ref. 22 (Table 1; cohorts G07–G11). DO mice were housed in pressurized, individually ventilated cages, provided with unrestricted access to acidified water and kept in temperature-controlled rooms maintained at  $21 \pm 1^\circ\text{C}$ . They were all on an ad libitum diet consisting of standard rodent chow SKOG from LabDiet.

### Blood variables, metabolic indices and mortality assessment

An ADVIA 2120i Hematology System (Siemens Healthineers) was used to quantitatively measure blood variables<sup>19</sup>. The hematologic profile and FBG were assessed every 3 months for each animal. In brief, mice were fasted for 6 h from 8:00 to 14:00, and blood was drawn by puncture of the submandibular vein. Blood glucose levels were measured using Contour test strips and a glucose meter (Ascencia Diabetes Care). For ADVIA measurements, 50  $\mu\text{l}$  of whole blood was resuspended in 150  $\mu\text{l}$  of RPMI-EDTA-K2 medium. The instrument was calibrated on the day of testing as per the manufacturer's instructions. BW was measured every 2 weeks at the same time of day under ad libitum conditions to avoid diurnal fluctuations.

All mice were followed until their natural death; however, one arm of the SLAM study involves establishing a tissue bank with specimens collected from various age groups<sup>19</sup>, and, therefore, a substantial portion of the animals was deliberately selected for euthanasia (376 males and 362 females; Supplementary Table 9), which was scheduled approximately every 6 months. For each cohort of 200 animals entering the study, where three or four animals were housed per cage, the selection process for euthanasia was entirely random; but the euthanasia procedure was no longer random when fewer than three animals were housed per cage, in which case euthanasia was preferentially directed toward single-housed animals over those housed in pairs and pairs over those housed in groups of three. This approach aimed to minimize any potential impact of single housing on mouse behavior, including factors such as feeding and grooming. Only animals that died of natural death or whose conditions were incompatible with life and were euthanized as per veterinarian guidance were accounted in survival analyses.

### FI

The protocol for the FI determination<sup>35</sup> was carried out by highly trained individuals at the NIA vivarium. First, the mice were transferred to the procedure room, where BW was recorded, and they were allowed an acclimatization period of 1 h. Subsequently, each mouse was individually assessed to determine multiple health deficits (Supplementary Table 10), taking approximately 5 min for each one. All deficits were assessed on a scale of 0, 0.5 or 1 by the evaluator. Body temperature was monitored using implantable RFID microchips from Bio Medic Data Systems.

### Statistics and reproducibility

**Software.** All analyses were performed using R version 4.3.0 and RStudio version 2023.03.0. +386. The full statistical analysis code is publicly available (see 'Data availability' section). The following R packages (version) were downloaded from the Comprehensive R Archive Network (CRAN) (R Foundation for Statistical Computing) and used to develop the pipeline, perform the analyses and generate the figures and tables:

abind (1.4.5), backports (1.4.1), bit (4.0.5), bit64 (4.0.5), bitops (1.0.7), broom (1.0.4), bslib (0.4.2), cachem (1.0.7), car (3.1.2), carData (3.0.5), cli (3.6.1), colorspace (2.1.0), compiler (4.3.0), crayon (1.5.2), curl (5.0.0), data.table (1.14.8), digest (0.6.31), dplyr (1.1.2), eha (2.10.3), evaluate (0.20), exact2x2 (1.6.8), exactci (1.4.2), extrafont (0.19), extrafontdb (1.0), farver (2.1.1), fansi (1.0.4), fastmap (1.1.1), forcats (1.0.0), generics (0.1.3), ggplot2 (3.4.2), ggpubr (0.6.0), glue (1.6.2), grid (4.3.0), gridExtra (2.3), groupdata2 (2.0.2), gtable (0.3.3), h2o (3.40.0.1), hms (1.1.3), htmltools (0.5.5), httr (1.4.5), jsonlite (1.8.4), jquerylib (0.1.4), kableExtra (1.3.4), knitr (1.42), km.ci (0.5.6), KMSurv (0.1.5), labeling (0.4.2), laplacesDemon (16.1.6), lattice (0.21.8), lifecycle (1.0.3), lme4 (1.1.33), lmerTest (3.1.3), lubridate (1.9.2), magrittr (2.0.3), MASS (7.3-58.4), Matrix (1.5.4), MatrixModels (0.5.1), Metrics (0.1.4), minqa (1.2.5), munsell (0.5.0), mgcv (1.8.42), nloptr (2.0.3), nlme (3.1.162), numDeriv (2016.8-1.1), parallel (4.3.0), pillar (1.9.0), pkgconfig (2.0.3), purrr (1.0.1), R6 (2.5.1), Rcpp (1.0.10), RCurl (1.98-1.12), readr (2.1.4), readODS (1.8.0), rio (0.5.29), rlang (1.1.1), rmarkdown (2.21), reshape2 (1.4.4), Rttf2pt1 (1.3.12), rvest (1.0.3), readxl (1.4.2), sass (0.4.5), scales (1.2.1), spline

(0.5.6), stringi (1.7.12), stringr (1.5.0), svglite (2.1.1), survival (3.5-5), survminer (0.4.9), systemfonts (1.0.4), textshaping (0.3.6), tibble (3.2.1), tidyr (1.3.0), tidyselect (1.2.0), tidyverse (2.0.0), timechange (0.2.0), tools (4.3.0), utf8 (1.2.3), vctrs (0.6.2), vroom (1.6.3), viridisLite (0.4.1), withr (2.5.0), xfun (0.39), xtable (1.8-4), xml2 (1.3.4) and zoo (1.8-12).

**The hematology-based clock.** A machine learning pipeline was built to predict age. Given that chronological age was the primary outcome during the clock construction, we deliberately omitted age and other chronological variables, including wave (the ordinal visit number), cohort and study, from serving as predictors in the models before building the DNN.

An essential premise in the development of tools for biological age prediction is the removal of any chronological feature that alters the tool's true ability to predict age. Both the SLAM and JAX studies are cohort based by design, lending to the possibility of subpopulations being unbalanced in terms of outcome. This, in turn, could artificially increase clock performance even after batch harmonization. Indeed, the inclusion of the categorical variable 'batch' as a feature in the clock improved prediction by more than 5%. Initially represented as an ordinal label with no other purpose than to indicate the sequence in which each order of animals was delivered by the supplier, the variable 'batch' has, in fact, pertinent 'age-related' information for the clock. SLAM animals arrived at the NIA vivarium in cohorts of approximately 200 animals every 3 months, balanced in number between strains and sexes, for a total of 10 cohorts (C01–C10). Blood samples were typically collected every 3 months from baseline, but the coronavirus disease 2019 (COVID-19) pandemic restrictions caused some batches of SLAM mice to have missing observations at specific timepoints, leading to unbalanced representation of the outcome 'age' across batches. On the other hand, the JAX study with its five cohorts (G07–G11) had blood sample collection at only three timepoints, far fewer than SLAM. Hence, depending on the specific batch number and its associated age frequency distribution, the supposedly inert categorical 'batch' label was now providing an important chronological hint for the model to predict age and improve performance. Therefore, variables such as the blood test number (that is, the sequential visit number in epidemiological studies), the study (that is, SLAM versus JAX) and the batch label were excluded from the set of features used to predict age. This concept is applicable to any other categorical variable that is not well balanced between levels or classes in terms of the outcome representation.

**Data pre-processing.** Sample size details are indicated in Table 1. No statistical method was used to predetermine sample size; the experiments were not randomized; and the investigators were not blinded to allocation during experiments, outcome assessment or analysis. Pre-processing of biological variables (that is, hematological variables, BW and FBG) included data cleaning and transformation (logit for variables expressed as percentages and log for the rest), handling of missing data and batch effect harmonization. This process was carried out after splitting the dataset into the training (72%) and testing (28%) sets to avoid eventual information leakage between training and testing datasets. Data distribution was assumed to be normal, but this was not formally tested. Outliers were identified and removed based on the 1.5× interquartile range method stratified by sex and strain after variable log transformation. To rectify batch effects, we estimated the effect of the variable cohort on each feature fitting a linear mixed-effect model<sup>38</sup> through restricted maximum likelihood and subsequently subtracted the estimates (see Supplementary Table 11 for results in the training set). These models encompassed cohort and animal ID as random effects, whereas sex and strain were included as fixed effects (equation (1)).

$$\text{biomarker}_{ij} = \beta_{0j} + \beta_{1j} \cdot \text{Sex}_{ij} + \mathbf{b}_{0k} + \mathbf{b}_{1l} + \mathbf{b}_{2m} + \epsilon_{ij} \quad (1)$$

Where:

- **biomarker<sub>ij</sub>** represents the biological variable (that is, hematological variables, BW and FBG) for observation *i* within group *j* in where the batch effect shall be removed.
- $\beta_{0j}$  and  $\beta_{1j}$  are the fixed effect coefficients associated with the intercept and sex, respectively, for each group *j*.
- $\mathbf{b}_{0k}$ ,  $\mathbf{b}_{1l}$  and  $\mathbf{b}_{2m}$  are the random intercepts for each unique identifier mouse<sub>*k*</sub>, cohort<sub>*l*</sub> and strain<sub>*m*</sub>, respectively.
- $\epsilon_{ij}$  is the error term for observation *i* within group *j*.

In detail, the process encompasses fitting a linear mixed model (LMM) to each biological variable, extracting the random effect coefficient assigned to each cohort and each feature and then subtracting those coefficients from the raw value of each feature. To confirm batch effect removal, we fitted the same LMMs again to verify that the model was unable to detect any random effect assigned to the cohort variable after harmonization. Notably, we intentionally refrained from adjusting for age at this stage, as these features were intended for later incorporation into the predictive clock as explanatory variables for age prediction. Our aim was to prevent the use of any chronological information within the predictors before constructing the clock to prevent any artificial enhancement of model performance stemming from chronological aspects of the project.

In addressing the impact of individual evaluators during the harmonization of the FI variable, we followed a similar methodology including the evaluator, cohort, animal ID and age as random effects, with sex and strain serving as fixed effects. It is important to highlight that, in this instance, we corrected for age because FI was not intended for inclusion as an explanatory variable within the clock.

To determine eventual implication of missingness imputation on clock performance, three different imputation methods were applied—that is, (1) computing the longitudinal information of each mouse for a specific variable to fit a spline and using the model to predict the missing point (R package zoo<sup>39</sup>); (2) using a non-parametric random-forest-based algorithm (R package MissForest<sup>40</sup>); and (3) imputing the mean value for each specific timepoint, sex and strain. The variables PLTs and eosinophils were adjusted to clumps following the same approach described for batch effect correction.

All numerical features were standardized, and all categorical variables were binary transformed using one-hot encoding.

**Model selection.** A comparative analysis of three distinct models for blood age prediction was conducted using the H2O package<sup>21</sup>. These models included a DNN, a GBM and a GLM with elastic net penalization. We exclusively used the H2O package's default settings (Supplementary Table 12) while employing a 10-fold CV procedure on the training dataset. The model selection process involved choosing the one that exhibited the lowest MAE in the 10-fold CV. The selected model, which turned out to be the DNN (multi-layer feedforward artificial neural network employing stochastic gradient descent and back propagation), subsequently underwent a process of hyperparameter tuning, feature selection and final testing.

Several methods were applied to avoid overfitting of the hematological clock, including (1) the practice of placing all samples from the same mouse in one dataset; (2) ensuring a balanced representation of samples from all categorical variables in every subset; (3) performing the batch effect harmonization separately on each dataset; and (4) use of elastic net penalization as well as (5) early stopping and (6) dropout when training the DNN for the first and second hidden layers (50% dropout rate) and (7) imputing missingness separately on each dataset after splitting the sample.

**Feature importance and feature selection.** For the feature importance and feature selection processes, we initially included all the reportable parameters (referred to here as 'variables') from the ADVIA 2120/2120i



Hematology System as outlined on pages 1–22 of the operator's guide, except for user-definable Morphology Results because they may limit the generalized use of the clock. From this set of variables, those with pairwise correlations equal to or exceeding a threshold of  $r = 0.80$  were eliminated. The list that remained in this pairwise analysis included: RBC (excluding HGB and HCT), WBC (excluding #LYMPH), CHCM (excluding CHCMr), %LYMPH (excluding %NEUT), #RETIC (excluding %RETIC), %MONO (excluding #MONO), #NEUT (excluding %NEUT) and PLT (excluding PCT). Additionally, #BASO and %BASO were excluded due to the prevalence of more than 70% zero values. Lastly, the selection was further refined by excluding variables that did not overlap between JAX and SLAM datasets, resulting in a final compilation of 19 variables—that is, 15 blood variables (RBC, MCV, MCH, MCHC, CHCM, RDW, HDW, MPV, %LYM, #EOS, WBC, #NEUT, %EOS, %MONO and PLT), along with two metabolic indices, FBG and BW, and two demographic variables, SEX and STRAIN. From this pool of 19 features undergoing a process of feature selection, we ultimately arrived at a refined collection of 12 variables, which constitutes the set that stands as the core of the hematology-based clock—that is, RBC, MCV, MCH, MCHC, CHCM, HDW, WBC, PLT, FBG, BW, SEX and STRAIN. The feature selection process that we conducted is described below.

To incorporate longitudinal transitions of features into the model, new variables (delta) for each numeric variable were created, which captured the change (increase or decrease) between two consecutive timepoints. In cases where there was no previous timepoint available, the initial delta value was set to 0.

A feature selection process applying the methodology described in ref. 27 and implemented in the H2O package was applied through a 10-fold CV to determine the optimal number of variables to include in the hematological clock. The feature importance ranking generated by the neural network trained during the model comparison process (DNN maximal with 19 features) was used to perform a feature selection process and identify the minimal adequate model. In a systematic approach, different DNNs, each with a progressively reduced number of features, were trained. This process involved iteratively removing one feature ranked as the less important predictor, then two features and so on, until training a DNN with just one feature, the one with higher importance determined by the ranking provided by DNN maximal. Lowest MAE and highest R in the 10-fold CV process along with the fewest number of predictors were the criteria applied to select the final model (Table 2 in the manuscript). This entire process was executed using data exclusively from the training dataset (72% data). The same method described in ref. 27 was applied to determine feature importance. In this case, blood samples were processed from the entire dataset for general feature importance and categorized by tercile lifespan for longitudinal importance.

**Hyperparameter tuning.** Hyperparameter fine-tuning of the network through a grid search approach involved various architectural configurations, including different depths ranging from four to six layers and varying the number of neurons based on a multiplicative factor of the number of inputs (27 inputs after feature selection and adding the delta variables). Additionally, different levels of penalization parameters L1 and L2 (Supplementary Table 2) were considered. All these optimizations were performed using the training dataset and carried out within the context of a 10-fold CV process.

Through hyperparameter tuning, the optimized DNN was designed as a four-layer architecture. This configuration included 12 features, an input layer with 27 nodes, a first hidden layer with 108 neurons using a rectified linear unit (ReLU) activation function and a second hidden layer with 432 neurons also using the ReLU activation function, an output layer containing a single output with a linear activation function and penalization parameters  $L1 = 1 \times 10^{-4}$  and  $L2 = 1 \times 10^{-5}$ . The model's parameter count encompassed 50,004 weights and 541 biases, resulting in a cumulative total of 50,545 estimates.

Regarding the metrics for determining model performance after hyperparameter tuning, we opted for the Pearson's coefficient R over  $R^2$  as a metric for measuring correlation between predicted age determined by this final model and chronological age because, whereas both shed light on the linear relationship between predicted and actual values, R specifically indicates the direction of that relationship that proves valuable in understanding the relationship dynamics, especially when segmenting predictions. Additionally, R is beneficial for clock comparisons in the absence of other metrics. Our preference for MAE as a metric to compare models stems from the fact that the error distribution is not Gaussian, making RMSE a less appropriate metric<sup>41</sup>.

**Model validation.** After the construction of the neural network, age predictions were generated and validated using the held-out group (28% of the total) and the randomly selected cohort C07 G08 (10% of total data).

**Genetic heterogeneity and phenotype association analyses.** *Genetic heterogeneity and model performance.* We estimated the extent of heterogeneity between strains by evaluating how individual-level variability within each strain influences the overall variability observed on each biomarker and each strain. To accomplish this, we conducted an analysis to compute the ICC for each biomarker. This process entailed determining the ICC within each of the three strains as the ratio between the variance attributed to the animals after data harmonization (to mitigate batch effects) and the total variance (equation (2)). Initially, we applied a linear mixed-effects<sup>38</sup> model to each biomarker, accounting for both fixed effects (sex and age) and random effects (animal ID). Variance components necessary for ICC calculation were extracted from the variance–covariance matrix of these models. Subsequently, we compared this variability across strains using ICC ratios (for example, for RBCs and B6 versus HET3,  $ICC_{B6-RBC} / ICC_{HET3-RBC}$ ), which provides information about the relative influence of individual-level variability on each biomarker between pairs of strains.

$$ICC_{kj} = \frac{\text{Between Animal variance}_{kj}}{\text{Total variance}_{kj}} \quad (2)$$

Where  $k$  represents  $k$ -th biomarker analyzed, and  $j$  represents the  $j$ -th strain.

To maintain the longitudinal structure comparable, the SLAM dataset was filtered to include only the three overlapping timepoints present in the JAX study (blood samples collected at 12 months, 18 months and 24 months of age). To compare similar population sizes and avoid sample bias, multiple resampling (100i) composed of 70% of SLAM samples (approximately the same size as the JAX dataset) was performed, and the results were averaged. The calculations were done in harmonized datasets after batch effect removal.

**Aging gap.** Aging gap was calculated by taking the differences between predicted and chronological age.

**Aging gap and survivorship.** Due to the presence of heteroskedasticity in the aging gap variable, which could bias our survival analysis, we performed these analyses in two different scenarios: (1) considering all observations on each dataset and (2) limiting the data to observations collected when the animals were young or adult (up to 70 weeks) and removing all samples from those animals that died before this 70-week threshold. This second approach was done only for the animals whose samples were not included during model construction (held-out and C07 G08 sets). Then, aging gap was used as a prognostic factor to identify differences in survival risk through extended Cox and Gompertz regression models both as a continuous variable and as a categorical variable.

To investigate whether the slope of the aging gap trajectories had implications on mouse lifespan, a linear mixed regression was used to



extract the slope of the aging gap trajectory for each mouse (equation (3)). The slope distribution was then segregated into terciles, and the following two tests were subsequently performed. (1) Slope as a continuous variable. A linear regression of slope aging gap on lifespan adjusted by sex, strain and cohort. (2) Slope as a categorical variable. A comparison between the tercile 1 slope aging gap (labeled as slow agers) versus tercile 3 slope aging gap (labeled as fast agers) to identify which group had a larger proportion of individuals living beyond the P90 lifespan (144 weeks). This comparison was conducted using a combination of quantile regression and Boschloo's test, as previously described<sup>42</sup>. All these analyses were performed on animals with instance of natural death and blood draws in at least two timepoints that were included in the held-out group during the construction of the clock (28% data).

$$\text{AgingGap}_{ij} = \beta_0 + \beta_1 \times \text{Sex}_{ij} + \beta_2 \times \text{Age}_{ij} + b_{0i} + b_{1i} \times \text{Mouse} + b_{2i} \times \text{Cohort} + b_{3i} \times \text{Strain} + \epsilon_{ij} \quad (3)$$

Where:

- **AgingGap<sub>ij</sub>** is the aging gap for the *j*-th observation of the *i*-th individual.
- $\beta$  represents the intercept and the fixed effect coefficients.
- $b_{0i}$  is the random intercept for each mouse.
- $b_{1i}$  is the random slope within each mouse.
- $b_{2i}$  is the random intercept for each cohort.
- $b_{3i}$  is the random intercept for each strain.
- $\epsilon_{ij}$  is the residual error term for the *j*-th test of the *i*-th animal.

The aging gap was used as a prognostic factor to identify differences in survival risk through extended Cox and Gompertz regression models in two distinct approaches: (1) the aging gap was incorporated as a continuous time-dependent variable in an extended model (equations (4) and (6)), and (2) it was included as a categorical risk factor in an extended recurrent event model (equations (5) and (7)). A counting process dataset that accommodated multiple observations for each mouse and time on specific aging gap terciles was constructed by establishing intervals between blood tests for each animal spanning from the age of one blood test to the subsequent, with the final interval extending from the age of the last blood test to the point of death or censorship. This design enabled us to incorporate into the model the length (in weeks) that each animal spent in each aging gap tercile (and changes among terciles across life). Subsequently, both intervals and terciles were incorporated into a recurrent event model using both Cox and Gompertz regressions. All models were adjusted by age, sex, strain and cohort.

$$h(t | \mathbf{X}) = h_0(t) + \exp(\beta_1 \cdot \text{Aging gap}(t) + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Sex} + \beta_4 \cdot \text{Strain} + \beta_5 \cdot \text{Cohort}) \quad (4)$$

Where:

- $h(t|\mathbf{X})$  is the hazard function at time *t*, given covariates **X**.
- $h_0(t)$  is the baseline hazard function.
- $\beta$  represents the regression coefficients.
- **Aging gap**(*t*) represents the aging gap at time *t*, treated as a continuous time-dependent covariate.

$$h_j(t | \mathbf{X}, N_{j-1}(t)) = h_{0j}(t) \cdot \exp(\beta_1 \cdot \text{Aging gap}^{\text{Tercile}}(t) + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Sex} + \beta_4 \cdot \text{Strain} + \beta_5 \cdot \text{Cohort}) \quad (5)$$

Where:

- $h_j(t|\mathbf{X}, N_{j-1}(t))$  is the hazard function for the *j*-th event at time *t*, given covariates **X** and the history of previous events  $N_{j-1}(t)$ .
- $h_{0j}(t)$  is the baseline hazard function for the *j*-th event.
- $\beta$  represents the regression coefficients.

- **Aging gap**<sup>Tercile</sup>(*t*) represents the aging gap tercile at time *t*.

$$h(t|\mathbf{X}) = \lambda_0 \cdot \exp(\beta_1 \cdot \text{Aging gap}(t) + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Sex} + \beta_4 \cdot \text{Strain} + \beta_5 \cdot \text{Cohort}) \quad (6)$$

Where:

- $h(t|\mathbf{X})$  is the hazard function at time *t*, given covariates **X**.
- $\lambda_0$  is the baseline hazard rate.
- **Aging gap**(*t*) represents the aging gap at time *t*, treated as a continuous time-dependent covariate.

$$h_j(t | \mathbf{X}, N_{j-1}(t)) = \lambda_{0j} \cdot \exp(\beta_1 \cdot \text{Aging gap}^{\text{Tercile}}(t) + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Sex} + \beta_4 \cdot \text{Strain} + \beta_5 \cdot \text{Cohort}) \quad (7)$$

Where:

- $h_j(t|\mathbf{X}, N_{j-1}(t))$  is the hazard function for the *j*-th event at time *t*, given covariates **X** and the history of previous events  $N_{j-1}(t)$ .
- $\lambda_{0j}$  is the baseline hazard rate for the *j*-th event.
- **Aging gap**<sup>Tercile</sup>(*t*) represents the aging gap tercile at time *t*.

**Aging gap and frailty.** In our examination of the relationship between aging gap and FI, we began by filtering the SLAM dataset to include only animals with longitudinal blood test and FI test records. This resulted in a total of 562 *M. musculus* with blood measures and FI scores (278 females, 284 males, 239 B6 and 323 HET3) and 631 samples from 51.9 weeks to 143.4 weeks of age. The blood dataset and FI dataset were joined using the common identifiers of animal ID and the age at the test in months. In situations where the animal's age did not align precisely between the two datasets, the FI value from the previous month was assigned to the missing timepoint. First, a linear mixed-effect model was performed to regress FI-corrected (harmonized for evaluator and cohort effects, as described in the pre-processing section) on aging gap, sex, strain and age (fixed effects) while incorporating animal ID as a random effect (equation (8)). To confirm the results of the previous analysis, we then performed a second approach. FI-corrected was regressed on aging gap using a piecewise mixed model and introducing a single knot at FI-corrected equal to 0.90. To determine the best segmentation point, all unique FI-corrected were ranked, leaving out the tail points in the rank, and all available models were fitted iteratively after segmenting on each FI-corrected. Finally, the two-segment piecewise model that minimized the residual standard error was selected. All models were adjusted by sex, strain and age as fixed effects and cohort and animal ID as random effects. The distribution of the FI-corrected was explored graphically and assumed to be normal.

$$\text{FICorrected}_{ij} = \beta_0 + \beta_1 \times \text{Aging gap}_{ij} + \beta_2 \times \text{Sex} + \beta_3 \times \text{Strain} + \beta_4 \times \text{Age}_{ij} + b_{0i} + \epsilon_{ij} \quad (8)$$

Where:

- **FICorrected<sub>ij</sub>** is the corrected FI (eliminating evaluator and cohort effect from each raw FI) for the *j*-th test of the *i*-th animal.
- $\beta$  represents the intercept and the fixed effect coefficients.
- $b_{0i}$  is the random intercept for each animal, capturing individual variations.
- $\epsilon_{ij}$  is the residual error term for the *j*-th test of the *i*-th animal.

## Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

All murine data are available from the lead contact upon reasonable request, as this is an ongoing cohort study in which animals continue to be enrolled.

## Code availability

All custom-written scripts necessary to use our clock and reproduce the results have been made publicly available and can be found on the GitHub repository at [https://github.com/jqmarron/0014\\_hematological\\_clock\\_2023\\_Git](https://github.com/jqmarron/0014_hematological_clock_2023_Git). Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon reasonable request.

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## Author contributions

J.M.-R. conducted data analysis, wrote the main codes and created the figures. W.M., M.E.F., J.C., O.M., Y.-H.H., Z.L., N.A., E.J.S. and L.J.L. contributed to establishing epidemiological and statistical methods. C.S. wrote the code for merging the datasets. S.C., N.L.P., C.V.L.-T. and D.P. contributed to the SLAM study supervision. J.M.-R., M.B., N.L.P. and E.J.S. wrote the original draft of the manuscript. J.M.-R., M.E.F., M.B., N.L.P. and R.d.C. reviewed and edited subsequent versions of the manuscript. M.B. created the figures. G.A.C. supervised JAX mouse studies and facilitated data collection. A.D. curated the data from the JAX study. J.M.-R. and M.E.F. curated data from the SLAM study. D.B.A., S.D., C.S. and N.E.-V. reviewed portions of the analytic code and output and advised on statistical methods. L.F. and R.d.C. conceived and supervised the SLAM study. All authors read and approved the final version of the manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

**Extended data** is available for this paper at <https://doi.org/10.1038/s43587-024-00728-7>.

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s43587-024-00728-7>.

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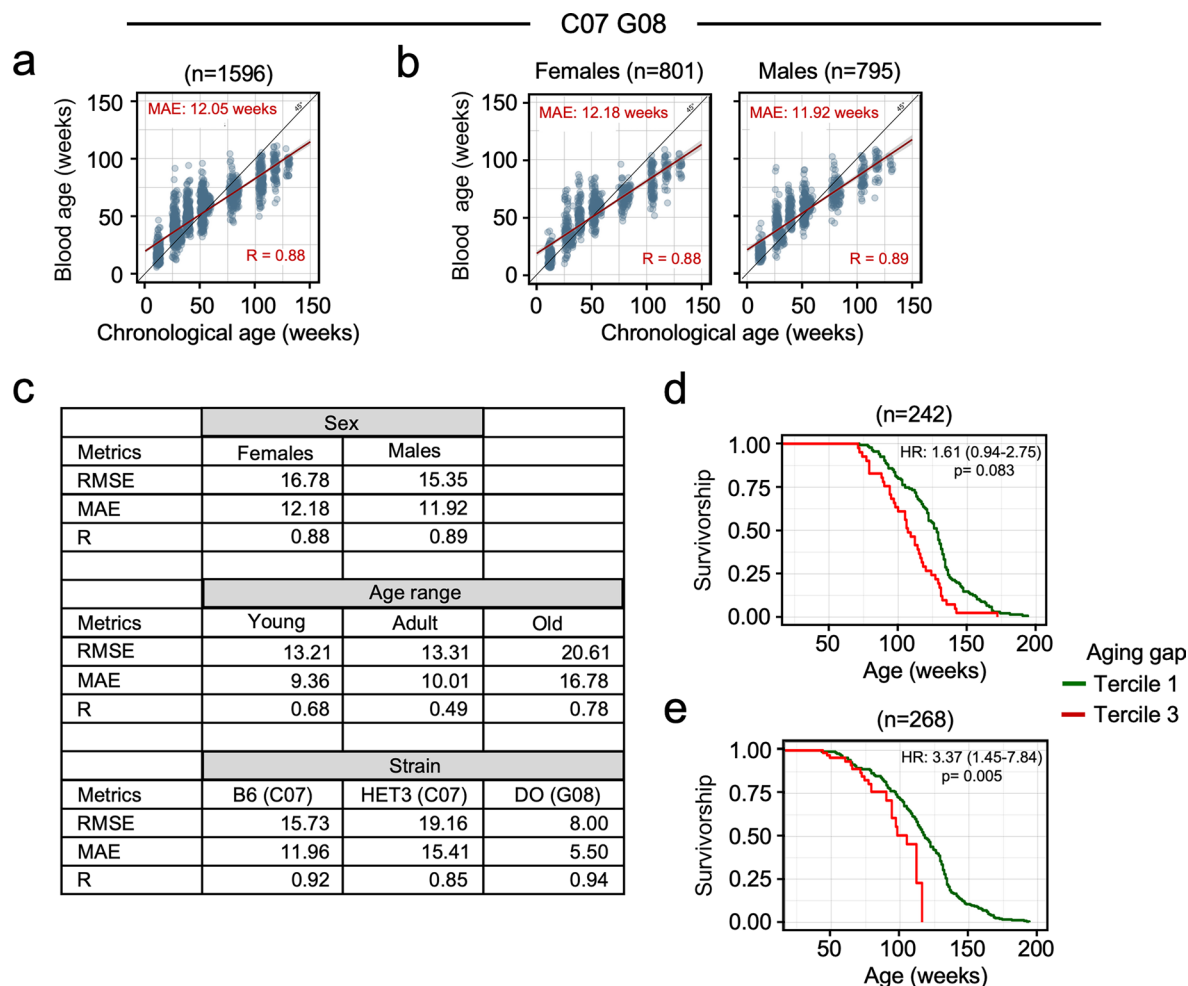
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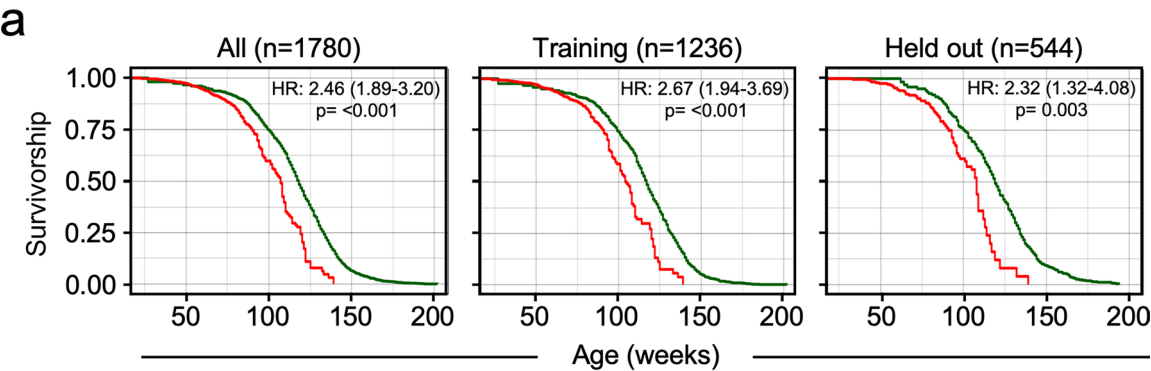
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**Extended Data Fig. 1 | Age prediction results and survivorship analyses in the randomly selected cohorts C07 and G08.** **a**, Result of clock's predictions after processing samples from the cohort C07 G08. n, number of observations. The red line shows an Ordinary Least Squares (OLS) regression, with the shaded grey area indicating the 95% confidence interval. Blue dots represent individual observations R, Pearson's correlation coefficient; MAE, mean absolute error. **b**, Result of clock's predictions stratified by sex. MAE, mean absolute error. (a and b) The red lines represent the Pearson correlation coefficient, the shaded grey areas the confidence intervals at 95%, and the blue dots represent individual observations. **c**, Impact of sex, age, and strain on the hematological clock age predictions. Young, <52 weeks of age; Adult, 52-82 weeks; Old, >82 weeks. B6, C57BL/6J; HET3, UM-HET3, DO, Diversity Outbred. RMSE, root mean squared

error. **a**, **b** and **c**, 357 mice, *Mus musculus* [177 Females, 180 Males, 116 DO, 119 B6 and 122 HET3], and 1,596 samples from 11.6 to 130.4 weeks of age. **d**, Kaplan-Meier curves depict mice categorized as fast agers (Tercile 3 aging gap) and slow agers (Tercile 1 aging gap) in the C07 G08 cohort, processing samples within the 70-week-window from animals alive after 70 weeks (242 mice, *Mus musculus* [117 Females, 125 Males, 95 DO, 74 B6 and 73 HET3], and 759 samples from 11.6 to 70.0 weeks of age). Hazard ratios (HR) were estimated using recurrent extended Cox models (two-sided). No adjustments were made for multiple comparisons. **e**, Similar analysis to (d) was carried out using all samples from the C07 G08 population (268 mice, *Mus musculus* [130 Females, 138 Males, 104 DO, 84 B6 and 80 HET3], and 1,123 samples from 11.6 to 130.4 weeks of age).



b

| Held out                                         |                                 |            |                                 |            |          |      |
|--------------------------------------------------|---------------------------------|------------|---------------------------------|------------|----------|------|
| Lifespan<br>above 90 <sup>th</sup><br>percentile | Fast agers<br>(Tercile 3 slope) |            | Slow agers<br>(Tercile 1 slope) |            | p-value* | Sig. |
|                                                  | Mice (n)                        | Proportion | Mice (n)                        | Proportion |          |      |
| Yes                                              | 4                               | 2.4%       | 31                              | 18.6%      | <0.001   | ***  |
| No                                               | 162                             | 97.6%      | 136                             | 81.4%      |          |      |
|                                                  |                                 |            |                                 |            |          |      |
| Total                                            | 166                             | 100.0%     | 167                             | 100.0%     |          |      |

**Extended Data Fig. 2 | Predicting the impact of aging gap on survivorship.**  
**a**, Survival analysis comparing animals in the aging gap terciles T1 versus T3 conducted for the entire study (1,780 mice, *Mus musculus* [895 Females, 885 Males, 484 DO, 661 B6 and 635 HET3], and 8,278 samples from 11.6 to 150.9 weeks of age) (n = 1780), the training set (1,236 mice, *Mus musculus* [622 Females, 614 Males, 302 DO, 484 B6 and 450 HET3], and 5,857 samples from 12.0 to 150.9 weeks of age) (n = 1236), and the held-out set (544 mice, *Mus musculus* [273 Females, 271 Males, 182 DO, 177 B6 and 185 HET3], and 2,421 samples from 11.6 to 145.6 weeks of age) (n = 544). Only samples from animals that died of natural death

were included in this analysis. **b**, Results of the maximum lifespan comparison show the proportion of animals living beyond the 90th percentile lifespan in the fast and slow aging groups, based on the slope-aging-gap categorization in the held-out dataset. This analysis includes samples from animals with at least 2 timepoints that died of natural causes. (500 mice, *Mus musculus* [251 Females, 249 Males, 150 DO, 172 B6 and 178 HET3], and 2,353 samples from 11.6 to 145.6 weeks of age). The p-value was calculated using a two-sided Boschloo's test.  
\*\*\* P < 0.001.

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Software and code

Policy information about [availability of computer code](#)

|                 |                                                                                    |
|-----------------|------------------------------------------------------------------------------------|
| Data collection | <div>Software in the ADVIA 2120i Hematology System (Siemens) with Autoslide.</div> |
|-----------------|------------------------------------------------------------------------------------|



## Data analysis

All custom-written scripts necessary to use our clock and reproduce the results is publicly available at [https://github.com/jqmarron/0014\\_hematological\\_clock\\_2023\\_Git](https://github.com/jqmarron/0014_hematological_clock_2023_Git).

All analyses were performed using R version 4.3.0 and Rstudio version 2023.03.0. +386. The full statistical analysis code is publicly available, see "Data Availability". The following R packages (version) have been downloaded from the Comprehensive R Archive Network, (CRAN), R Foundation for Statistical Computing, and used to develop the pipeline, perform the analyses, and generate the figures and tables: abind (1.4-5), backports (1.4.1), bit (4.0.5), bit64 (4.0.5), bitops (1.0-7), broom (1.0.4), bslib (0.4.2), cachem (1.0.7), car (3.1-2), carData (3.0-5), cli (3.6.1), colorspace (2.1-0), compiler (4.3.0), crayon (1.5.2), curl (5.0.0), data.table (1.14.8), digest (0.6.31), dplyr (1.1.2), eha (2.10.3), evaluate (0.20), exact2x2 (1.6.8), exactci (1.4-2), extrafont (0.19), extrafontdb (1.0), farver (2.1.1), fansi (1.0.4), fastmap (1.1.1), forcats (1.0.0), generics (0.1.3), ggplot2 (3.4.2), ggpubr (0.6.0), glue (1.6.2), grid (4.3.0), gridExtra (2.3), groupdata2 (2.0.2), gtable (0.3.3), h2o (3.40.0.1), hms (1.1.3), htmltools (0.5.5), httr (1.4.5), jsonlite (1.8.4), jquerylib (0.1.4), kableExtra (1.3.4), knitr (1.42), km.ci (0.5-6), KMSurv (0.1-5), labeling (0.4.2), laplacesDemon (16.1.6), lattice (0.21-8), lifecycle (1.0.3), lme4 (1.1-33), lmerTest (3.1-3), lubridate (1.9.2), magrittr (2.0.3), MASS (7.3-58.4), Matrix (1.5-4), MatrixModels (0.5-1), Metrics (0.1.4), minqa (1.2.5), munsell (0.5.0), mgcv (1.8-42), nloptr (2.0.3), nlme (3.1-162), numDeriv (2016.8-1.1), parallel (4.3.0), pillar (1.9.0), pkgconfig (2.0.3), purrr (1.0.1), R6 (2.5.1), Rcpp (1.0.10), RCurl (1.98-1.12), readr (2.1.4), readODS (1.8.0), rio (0.5.29), rlang (1.1.1), rmarkdown (2.21), reshape2 (1.4.4), Rttf2pt1 (1.3.12), rvest (1.0.3), readxl (1.4.2), sass (0.4.5), scales (1.2.1), spline (0.5.6), stringi (1.7.12), stringr (1.5.0), svglite (2.1.1), survival (3.5-5), survminer (0.4.9), systemfonts (1.0.4), textshaping (0.3.6), tibble (3.2.1), tidyr (1.3.0), tidyselect (1.2.0), tidyverse (2.0.0), timechange (0.2.0), tools (4.3.0), utf8 (1.2.3), vctrs (0.6.2), vroom (1.6.3), viridisLite (0.4.1), withr (2.5.0), xfun (0.39), xtable (1.8-4), xml2 (1.3.4), and zoo (1.8-12).

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All murine data is available from the lead contact upon request, as this is an ongoing cohort study in which animals continue to be enrolled.

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## Life sciences study design

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Sample size

12,010 observations from 2,562 mice (*Mus musculus*). Sample size details are indicated in Table 1 of the manuscript. No statistical method was used to predetermine sample size. We use all the data collected in cohorts C01-C10 of the Study of Longitudinal Aging in Mice (SLAM) and cohorts G07-G11 of the The Jackson Laboratory's longitudinal study of aging (JAX). Moreover, as our dataset is the only one of its kind in terms of sample size (i.e., greater sample size than any other previously published study), we have taken steps to select our validation data in ways that ensure a fair evaluation of the model, and we have used several techniques to mitigate overfitting.

Data exclusions

Outliers for the biological variables (hematological variables, body weight and fasting blood glucose) were identified and removed based on

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data exclusions | the 1.5x interquartile range method stratified by sex and strain after variable log transformation. Additionally, because the aim of some analyses was to assess the association between the aging gap estimated by the hematological clock and mortality or the frailty index, animals that exhibited non-natural deaths or animals for which there was no frailty index data available may have been excluded. Similarly, animals that died before the 70-week threshold were specifically excluded from analysis that assess the potential impact of survival bias. All corresponding details have been reported in the manuscript. |
| Replication     | Analyses include two independent longitudinal studies (SLAM and JAX), each one of them with several cohorts (15 in total). Moreover, the whole dataset has been split in training, testing and held-out subsets. All relevant details have been reported in the manuscript                                                                                                                                                                                                                                                                                                                                                             |
| Randomization   | This is an observational study, so the potential confounders are adjusted in the regression models. Moreover, all splitting procedures on the datasets were performed randomly. Details have been reported in the manuscript.                                                                                                                                                                                                                                                                                                                                                                                                          |
| Blinding        | This is an observational study, so blinding is not relevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|                         |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Laboratory animals      | Female and male mice ( <i>Mus musculus</i> ), aged 11.6 to 150.9 weeks, of three strains (C57BL/6J, UM-HET3 and Diversity Outbred).                                                                                                                                                                                                                                                                                           |
| Wild animals            | N/A                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Reporting on sex        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Field-collected samples | N/A                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Ethics oversight        | Mice from The Study of Longitudinal Aging in Mice was approved by the Animal Care and Use Committee of the National Institute on Aging (Baltimore, MD) under Animal Study Protocol number TGB-458-2022. Mice from the Diversity Outbred (DO) study were reviewed and approved by The Jackson Laboratory's Institutional Animal Care and Use Committee (IACUC) under protocol #IACUC-06005, Jackson Center for Aging Research. |

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## Plants

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